powerful information technology was taboo. Computers were therefore given mainly to Soviet factory managers, and even they had to send all their data back to Moscow to be analyzed. As a result, Moscow was flooded with paperwork. By the 1980s, this unwieldy system of computers was producing 800 billion documents per year, all destined for the capital.[52]

However, at a time when IBM and the Soviet government declined to develop the personal computer, hobbyists like the members of the California Homebrew Computer Club resolved to do it by themselves. It was a conscious ideological decision, influenced by the 1960s counterculture with its anarchist ideas of power to the people and libertarian distrust of governments and big corporations.[53]

Leading members of the Homebrew Computer Club, like Steve Jobs and Steve Wozniak, had big dreams but little money and didn't have access to the resources of either corporate America or the government apparatus. Jobs and Wozniak sold their personal possessions, like Jobs's Volkswagen, to finance the creation of the first Apple computer. It was because of such personal decisions, rather than because of the inevitable decree of the goddess of technology, that by 1977 individuals could buy the Apple II personal computer for a price of $1,298 – a considerable sum, but within reach of middle-class customers.[54]

We can easily imagine an alternative history. Suppose humanity in the 1970s had access to the same scientific knowledge and technical skills, but McCarthyism had killed the 1960s counterculture and established an American totalitarian regime that mirrored the Soviet system. Would we have personal computers today? Of course, personal computers might still have emerged in a different time and place. But in history, time and place are crucial, and no two moments are the same. It matters a great deal that America was colonised by the Spaniards in the 1490s rather than by the Ottomans in the 1520s, or that the atom bomb was developed by the Americans in 1945 rather than by the Germans in 1942. Similarly, there would have been significant political, economic and cultural consequences if the personal computer emerged not in San Francisco in the 1970s but rather in Osaka in the 1980s or in Shanghai in the first decade of the twenty-first century.

The same is true of the technologies being currently developed. Engineers working for authoritarian governments and ruthless corporations could develop new tools to empower the central authority, by monitoring